



Intelligent leaf disease diagnosis: image algorithms using Swin Transformer and federated learning

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Accepted: 13 October 2024 / Published online: 7 November 2024

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Abstract

In recent years, the international population has surged which has led to increased pressure on food security. Plant diseases are one of the most essential factors affecting food security. Plant diseases often manifest pathological features on plant leaves. This paper combines deep learning with continual learning and federated learning, proposing a federated continual learning comprehensive model based on the Swin Transformer model (SSPW224-LwF-3). Our datasets come from shared datasets and integrate data from other datasets based on a ratio principle of 6:1:3 for the training set, validation set and test set (https://osf.io/v4qfr/?view_only=38b53e39988c4e1e82363031a921f799). Experimental results show that the SSPW224 model achieves an accuracy, precision, recall and F1-score of 97.20%, 95.25%, 94.85% and 94.71% on the test set. It is evident to find out that the SSPW224 model outperforms the other network models in the identification and classification of plant leaf diseases from the data and visualization perspectives. The average accuracy of the SSPW224-LwF-3 model strategy combination proposed in this study reaches the highest 47.06% among 18 combination strategies, with an accuracy of 49.41% and 33.81% for old and new data. The overall performance of this model is optimal. Therefore, the proposed SSPW224-LwF-3 model can achieve efficient, sustainable, and distributed recognition and classification of plant leaf diseases which provides a tool for plant disease identification in smart agriculture.

Keywords Leaf disease · Deep learning · Swin Transformer · Continual learning · Federated learning · *t*-SNE

1 Introduction

With the emergence of academic disciplines such as plant pathology and microbiology, research on plant diseases is gradually deepening. Leaf disease is one of the most common types of plant disease, which is caused by a certain plant disease and results in some aberrant leaf traits such as yellowing [1], black spots [2] or mosaic patterns [3] on the leaves. The etiology of leaf diseases encompasses multifaceted factors, such as fungal infection, bacterial infection, viral invasion and other causes such as mosquito bites.

The first etiological category is fungal leaf diseases [4], which is caused by fungi such as powdery mildew, rust disease and anthracnose. Exemplified by *Diplocarpon Rosae*

Wolf [5], a fungus that causes black spot disease in plants such as roses (the fungi that cause leaf black spot disease in different plants are not the same), *Diplocarpon Rosae* will form haustorium after adsorbing on host plants. This enables *Diplocarpon Rosae* to penetrate and intrude into the host plant's surface cells to obtain cellular nutrients for its growth. The host cells will gradually die after being invaded. After being infected, the host plant leaves progress from minute lesions to irregular blotches, and the color of the spots will gradually deepen, turning from brown to black-brown.

The second etiological category is bacterial leaf diseases [6], which is caused by bacterial infections such as leaf spot disease or bacterial wilt disease, and ultimately lead to abnormal symptoms such as wilt and necrosis on plant leaves. Illustratively, we take *Ralstonia solanacearum* [7] as an example. *Ralstonia solanacearum* invades the plant via the root system and infects it upwards through the plant's root system. Once it reaches the plant's vascular system, *Ralstonia solanacearum* will secrete a substance which is called "protein extracellular polymers (PEL)" to block the plant's vascular system. Accumulation of these polymers within the

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vascular bundles results in blockages that prevent normal water and nutrient ascent, ultimately leading to symptoms of wilting and necrosis on plant leaves.

The third etiological category is viral leaf diseases [8], which is caused by viral such as potato leafroll virus (PLRV) and tobacco mosaic virus (TMV). These viruses can be transmitted through insects or other diseased plants. We take tobacco mosaic virus (TMV) [9] as an example. Illustratively, TMV is an RNA virus which has had to use substances in host cells to achieve transcription and translation to duplicate themselves and then produce new viruses. These newly synthesized virus particles gather inside the cells and form elongated nanotube structures through self-assembly. This can disrupt the cell membrane of host plant cells, perturbing plant hormone regulation. Eventually, it elicits abnormal leaf characteristics such as yellowing, curling and disease spots on the leaves.

Leaf diseases may have a variety of effects on plants because leaf disease can hinder the photosynthesis of most plants to a certain extent. The ability to carry out photosynthesis normally is the key to whether plants can grow and survive normally. And leaf is the main place for plant photosynthesis. Leaf diseases can lead to a decrease in the number of cells which can carry out photosynthesis in plant leaves and a decrease in chlorophyll content in plant leaves. So, it will lead to some abnormal symptoms such as leaf yellowing and curling, which reduces the efficiency and effective area of plant photosynthesis. Plants cannot synthesize enough organic matter without enough photosynthesis, ultimately leading to serious consequences such as slowing down plant growth and lowering seed quality. If this malignant outcome is reflected significantly in the field of grain crops, it will lead to a substantial reduction in the yield of grain crops. The yield of grain crops is of vital significance to human survival and development and the harmonious development of society. Firstly, enough food is the key to food security and the basis for human survival and development. Secondly, food production is the economic pillar of many developing countries. The identification and classification of plant leaf diseases are the basis of studying and solving plant leaf diseases. Therefore, it is very important to improve the accuracy of identifying and classifying plant diseases.

In the early research of plant leaf diseases, manual feature extraction [10, 11] is a common method. Early research on plant leaf diseases primarily relied on manual feature extraction methods, such as visual observation and manual measurement of leaf color, shape and size.

However, the method of manual feature extraction for plant leaves has subjectivity, which is not sensitive enough to subtle disease symptoms and has limitations. Additionally, this method also has the disadvantages of being time-consuming and requiring a large amount of work. It is difficult for this method to cope with high-capacity datasets. With

the advancement of science and technology, people began to pursue the task of identifying the characteristics of leaves with high efficiency and high accuracy, so the use of machine learning in artificial intelligence to identify and classify plant leaf diseases came into being [12]. Prabhjot Kaur et al. used logistic regression techniques to sample collected features and employed a hybrid convolutional neural network to identify leaf disease, ultimately achieving an accuracy of 98.7%. Panuwat Mekha et al. [13] used a random forest classification algorithm to identify and categorize a dataset of rice leaf brown spot disease (BSR), finally achieving an accuracy of 69.44%. Ms. Deepika Chauhan et al. extensively employed artificial intelligence methods, such as naive Bayes (NB), decision trees (DT), K-nearest neighbors (K-NN), support vector machine (SVM) and random forest (RF), to detect crop diseases. Then they proposed a model which is the most suitable for efficient disease detection. The accuracy of the random forest model is the highest, at 80.68% [14]. Muhammad Alyas Khan [15] proposed using gray-level co-occurrence matrix (GLCM) and Color Moments to extract color and shape features of leaf segmentation regions. Finally, they put these extracted features into a backpropagation neural network (BPNN) classifier to detect and classify walnut leaf diseases. The experimental result shows that the detection accuracy of the proposed method is 95.3%. Marjanul Islam Tarik et al. [16] proposed an automated system which is based on image processing and machine learning for identifying and classifying potato leaf diseases. They analyzed more than 2034 images of unhealthy potatoes and healthy leaves from publicly accessible plant datasets. The program achieved a prediction accuracy of 99.23% when tested with 25% test data and 75% training data. Their result indicates that the method of machine learning surpasses all existing methods in the potato disease detection field. Sunil Harakannanavar et al. [17] used algorithms such as support vector machine (SVM), convolutional neural network (CNN) and K-nearest neighbors (K-NN) to extract tomato features and classify them. The accuracy of the proposed model was tested on diseased tomato samples by SVM, K-NN and CNN, and the accuracy was 88%, 97% and 99.6%, respectively. Muhammad Anwarul Azim et al. [18] used an extreme gradient boosting decision tree ensemble method on UCI's rice leaf disease dataset, resulting in an 86.58% accuracy. Seyed Mohamad Javidan et al. [19] achieved automatically separated diseased areas from healthy parts of leaves by using K-means clustering and extracted features in RGB, HSV and Lab color models. Harikumar Pallathadka et al. [20] proposed a framework based on machine learning for rice leaf disease classification and detection. SVM, naive Bayes and CNN are used in this framework. PCA algorithm was also used for feature extraction. Sambasivam et al. [21] proposed the use of category weights, SMOTE (synthetic minority oversampling

technique) and deep convolutional neural networks to combat high category imbalance with an accuracy of over 93%. These machine learning algorithms and technologies provide more possibilities for early diagnosis of plant leaf diseases, which are of great significance for maintaining food security. They also set the stage for the development of deep learning in the field of plant leaf diseases.

Deep learning is an end-to-end learning approach that allows tasks to be learned directly from raw data through end-to-end learning, and weights can be automatically adjusted during the training process. So, it can reduce the need for manual design processes. Compared to machine learning, deep learning models can often achieve better performance when dealing with large-scale and complex image data [22]. In the field of plant leaf disease recognition, Ümit Atila et al. proposed the framework of EfficientNet deep learning for plant leaf disease classification, and they trained the model on the PlantVillage dataset. The accuracy of EfficientNet-B5 and EfficientNet-B4 models in the original and enhanced datasets is 99.91% and 99.97%, respectively. Rakesh Chandra Joshi et al. proposed a method named VirLeafNet based on deep learning and achieved automatic virus infection detection for *Vigna mungo*. All proposed models achieved over 90% accuracy [6]. Andrew J. et al. [23] fine-tuned popular pre-trained models' hyperparameters. They used the PlantVillage dataset in experiments, and the DenseNet-121 model achieved a higher classification accuracy of 99.81%. B. Sai Reddy et al. [24] used convolutional neural networks (CNN) to train models with supervised pixel values, achieving an accuracy of over 97%. Shivali Amit Wagle [25] used some deep learning models such as AlexNet, VGG16, GoogLeNet, MobileNetv2 and SqueezeNet to classify and judge tomato plant leaves. Among them, the VGG16 model has the best performance, with an accuracy of 99.17%. Arun Malik et al. [26] proposed a mixed model combining VGG16 and MobileNet for sunflower disease identification and classification. On the same dataset, this method outperformed other algorithms with an accuracy of 89.2%. Hesham Tarek et al. [27] investigated various deep convolutional neural network (CNN) technologies such as ResNet50, InceptionV3, AlexNet and MobileNetV1 to detecting and identifying tomato leaf diseases. MobileNetV3 Large got the highest accuracy of 99.81% among other deep learning models, with a loss value of 0.0088. Yousef Methkal Abd Algani et al. [28] proposed a deep learning algorithm called ant colony optimization with convolutional neural network (ACO-CNN). They studied the diagnostic effect of plant leaf diseases by using ant colony optimization (ACO) and used a CNN classifier to extract image information features. Compared to existing techniques, the proposed method performs better in terms of accuracy. Marriam Nawaz et al. [29] proposed a method used for classifying tomato plant leaf diseases which

is based on ResNet-34 Faster-RCNN. ResNet-34 and convolutional block attention module (CBAM) were introduced as the feature extraction module for Faster-RCNN to extract deep-level key points. They tested this model on the PlantVillage Kaggle dataset and achieved an accuracy of 99.97%. Waleed Albattah et al. used an improved CenterNet to detect and classify various plant leaf diseases. In the experiment, they used DenseNet-77 for deep-level key point extraction. From this, it is evident that using deep learning methods to identify and classify plant leaf diseases has overcome the limitations of manually extracting image features compared to traditional methods and machine learning methods. And deep learning methods do not require manual intervention [30]. This makes the models more adaptable and can handle different types and degrees of diseases, which plays an important role in preventing and managing plant diseases.

However, in the classification process, each model has certain shortcomings that may not be applicable to some specific datasets. Catastrophic forgetting [31–33] is a serious but normal phenomenon that occurs in the process of deep learning when we solve practical problems. Catastrophic forgetting refers to the phenomenon where a neural network forgets previously learned tasks when learning new tasks. When a neural network is designed to solve multiple tasks, its weight parameters will be adjusted during the learning stage of new tasks, causing the loss of previously acquired knowledge. This phenomenon occurs because the weight parameters of a neural network inevitably change when learning new tasks to adapt to new inputs and objectives. The updates of these weights may lead to a decline in the network's performance on previous tasks, because the new parameter settings may no longer be suitable for the old tasks. Therefore, giving deep learning models the ability for continuous learning is the top priority in solving practical problems. Our research builds upon existing models and incorporate the strategies for federated learning and continual learning. The method we built gets more accuracy. And it is more efficient and resilient to the problem of existing catastrophic forgetting compared to other methods. The main contributions of this study are as follows:

- We have established a comprehensive and integrated datasets of various plant diseases images for training neural networks which is capable of identifying multiple plant diseases. We have also appropriately allocated the training sets, validation sets and test sets.
- We have compared several advanced deep learning methods on the comprehensive datasets we proposed and identified the model with the best performance on our datasets. We have validated the neural network based on the Swin Transformer architecture using multiple evaluation metrics for each classification. We applied the confusion matrix and *t*-SNE dimensionality reduction

methods to achieve visualized operation of evaluation metrics. We validate the superiority of the Swin Transformer neural network in the field of classification by data analysis and visual aspects.

- We have proposed a method which combines the Swin Transformer architecture with federated learning methods and various incremental continual learning methods. We integrate the fine-tuning, LwF and EWC continual learning methods with the Swin Transformer model so that we make the basic model get the ability of incremental learning. We also integrated federated learning methods with the Swin Transformer model in order to enhance the original model's ability for distributed learning.

This paper is organized into six parts. We provide an overview of the datasets and the preprocessing operations performed on the data in the second part. We describe the methods we used in these experiments and introduce the performance metrics in the third part. The fourth part presents the experimental results of our proposed method and other methods. The fifth part discusses our proposed method. Finally, we provide the conclusion of this paper in the sixth part.

2 Materials

2.1 Dataset

The dataset used in this paper is obtained from the Kaggle website. We chose a dataset named as Leaf Disease Dataset (combination) which focuses on plant leaf diseases. This dataset is based on many famous datasets such as the Plantvillage dataset, Rice Leaf Disease dataset and PlantifyDr dataset. Upon our review, we found errors in the cassava dataset in the original data, and the data for the apple cedar rust disease category were insufficient. Additionally, it was difficult to find enough suitable image data online to supplement this category. Therefore, we deleted the cassava dataset and apple cedar rust from this dataset. The basic quantity information of the dataset is shown in Table 1. Our dataset is divided into training sets, testing sets and validation sets. All of the pictures are well organized in the training directory, testing directory and validation directory within their relevant class name folders. We optimized the quantity distribution of picture data in the training sets, validation sets and testing sets based on the original dataset. The final ratio of picture data for each disease subclasses (including “healthy”) in the training sets, validation sets and testing sets is 6:1:3. Then, to avoid overfitting of the model, we increased the number of image data of the original dataset according to the set ratio. We have endeavored to ensure an ample number of samples in the test set, with the majority of test set samples exceeding

200 in quantity. The dataset includes 12 plant classes, further subdivided into 38 plant disease subclasses (including “healthy”). They are all standardized to 256×256 size and RGB color scheme.

We take the five leaf diseases of tomatoes as an example. Figure 1 shows the illustrations of healthy tomato leaves and other leaves who suffer from these five diseases: tomato mosaic virus, tomato yellow leaf curl virus, bacterial spot, early blight and leaf mold.

It is not difficult to see that the leaf of healthy tomato is uniformly green, with relatively few lobes and no significant differences in size between lobes. The leaf margins are smooth. Tomato leaves infected with tomato mosaic virus (TMV) will show color differences on the surface. It is obvious that there are variations in green color intensity at different positions on the leaves. There will be a slight curvature phenomenon at the leaf margin. Tomato leaves infected with tomato yellow leaf curl virus will exhibit severe curvature of the leaf margins and yellowing of the leaves. Tomato leaves infected with bacterial spot will show irregular dark brown spots which will continuously spread. Tomato leaves infected with early blight will have irregular black–brown spots and yellowing around the spots, too. And there will be a slight curvature phenomenon at the leaf margin. Tomato leaves infected with leaf mold will show a relatively concentrated yellowing phenomenon.

2.2 Data preprocessing

In order to enhance the diversity of data, the ability of generalization and performance of the model, it is necessary to preprocess the experimental data. We uniformly scale the images in the dataset to 224×224 pixels and apply blur processing to them. Then we normalize the image data. We also added salt-and-pepper noise and Gaussian noise to the images in the dataset to simulate random interference and scene changes in our real life. In order to improve the quality of the dataset, we perform data augmentation on the images in the training set, including random rotation, horizontal flipping, random cropping and randomly changing the brightness, contrast, saturation and hue attributes of the images. Taking an image from Apple Scale as an example, the following are the images of a plant leaf affected by Apple Scale disease that has undergone cropping, rotation, horizontal flipping, changing image attributes, blurring [34–36] and adding salt-and-pepper noise [37] or Gaussian noise [38].

Blurring is a common method during the process of data augmentation (see Fig. 2). It can improve the quality of our dataset and enhance the robustness of the model. Taking the example of blurring in preprocessing, blurring (or smoothing) is achieved by taking a weighted average of each pixel value with the weights of its surrounding pixels. For example, if we want to obtain image B after the blurring processing

Table 1 Original dataset sample data

Category	Disease	Train	Validation	Test	Total
Rice	Brown spot	324	52	147	523
	Healthy	893	148	447	1488
	Hispa	339	56	170	565
	Leaf blast	467	77	235	779
Apple	Apple scab	468	78	234	780
	Black rot	475	79	237	791
	Healthy	987	164	494	1645
Cherry	Healthy	512	85	257	854
	Powdery mildew	631	105	316	1052
Corn	Leaf spot	1182	197	591	1970
	Common rust	715	119	358	1192
	Healthy	697	116	349	1162
	Northern leaf blight	591	98	296	985
Grape	Black rot	708	118	354	1180
	Esca	822	138	423	1383
	Healthy	2054	387	935	3376
	Leaf blight	646	107	323	1076
Orange	HLB	3304	550	1653	5507
Peach	Bacterial spot	1378	229	690	2297
	Healthy	1233	205	617	2055
Bell pepper	Bacterial spot	598	99	300	997
	Healthy	887	147	444	1478
Potato	Early blight	600	100	300	1000
	Healthy	1055	176	527	1758
	Late blight	592	100	308	1000
Squash	Powdery mildew	1101	183	551	1835
Strawberry	Healthy	756	126	377	1259
	Leaf scorch	665	110	334	1109
Tomato	Bacterial spot	1276	212	639	2127
	Early blight	600	100	300	1000
	Healthy	1025	169	407	1691
	Late blight	1376	190	343	1909
	Leaf mold	577	95	280	952
	Mosaic virus	445	109	284	838
	Septoria leaf spot	1144	177	450	1771
	Spider mites	1076	167	433	1676
	Target spot	923	140	341	1404
	TYLCV	3539	535	1283	5357

of the original image A , we should firstly find out the image A and the blur kernel matrix K . Then, the result of blurring processing can be calculated by using formula $B = K * A$, where the symbol $*$ represents the convolution operation and K can be a matrix of various shapes and parameters. Taking Gaussian kernel matrix as an example, the Gaussian kernel matrix is obtained based on the following two-dimensional

Gaussian distribution function,

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (1)$$

This function represents the Gaussian distribution on a two-dimensional plane, and the Gaussian kernel matrix is obtained by sampling this function on discrete coordinates.

Fig. 1 Sample example of original data, **a** healthy, **b** tomato mosaic virus, **c** tomato yellow leaf curl virus, **d** bacterial spot, **e** early blight and **f** leaf mold

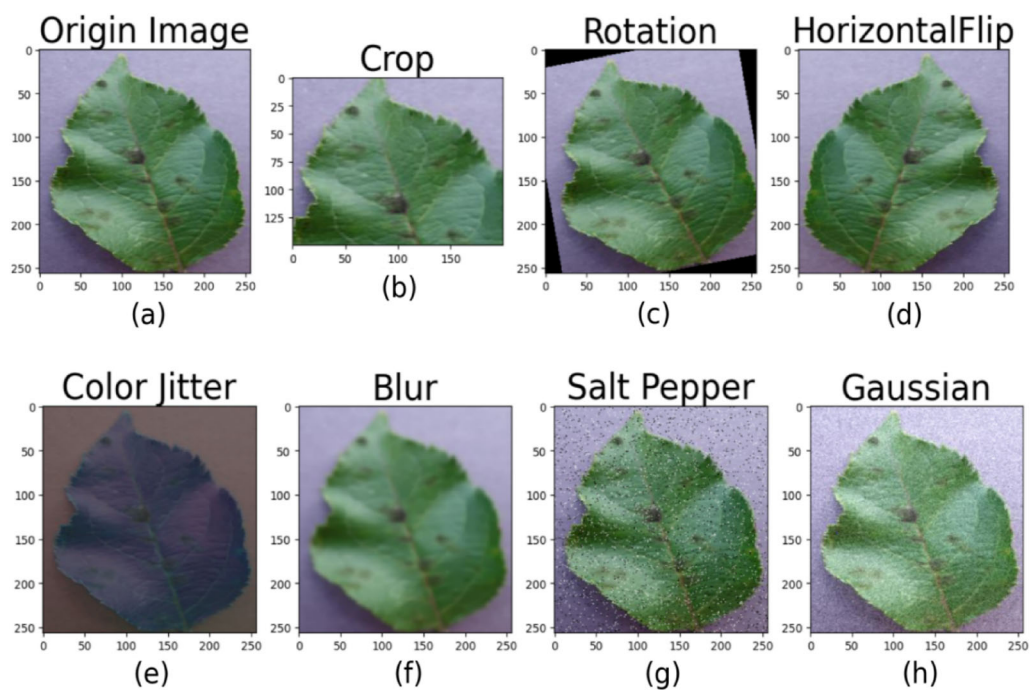
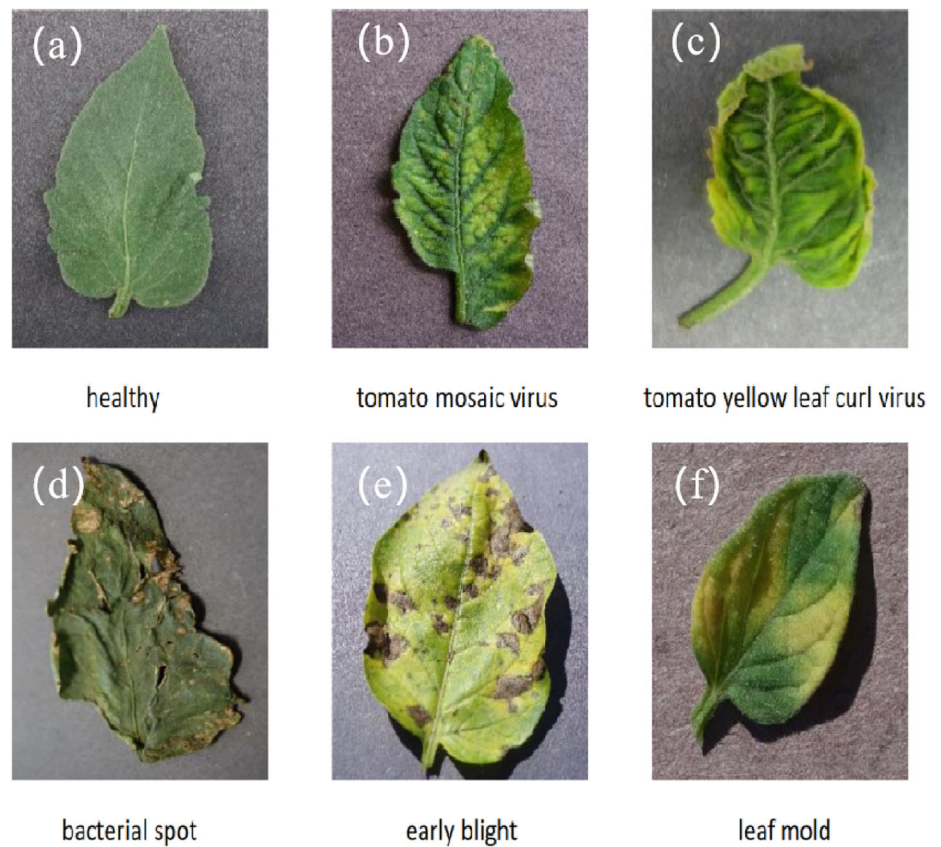


Fig. 2 Example of data preprocessing, **a** origin image, **b** crop, **c** rotation, **d** horizontal flip, **e** color jitter, **f** blur, **g** salt-and-pepper and **h** Gaussian

For example, when the standard deviation σ is 1, a simple 3×3 Gaussian kernel matrix can be sampled and calculated by using the above function and the result is as follows:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

3 Methods

3.1 Plant disease feature recognition model based on Swin Transformer architecture

Swin Transformer [39–42], proposed by Microsoft, is a deep learning model improved on the basis of the traditional Transformer model [39] and specializes in the process of visual data. Traditional Transformer architecture is composed of encoder and decoder, and is often used to process sequential data, such as natural language. Residual connection and layer normalization is between the input and output of each sub-layer. Each encoder layer includes a multi-head self-attention mechanism and a feed forward neural network, while self-attention mechanism is the core of Transformer architecture, which is used to calculate the value of positional encoding of each position in the sequence and consider the information of other ones. Positional encoding is often calculated by a combination of trigonometric functions:

$$\begin{aligned} PE_{(p, 2i)} &= \sin\left(\frac{p}{10000^{2i/d_{\text{model}}}}\right) \\ PE_{(p, 2i+1)} &= \cos\left(\frac{p}{10000^{2i/d_{\text{model}}}}\right) \end{aligned} \quad (2)$$

where p refers to position, i refers to the dimension index of the position encoding vector and d_{model} refers to the dimension of the model. After obtaining all information of the positions in the sequence, the information will go through a nonlinear transformation by the feedforward neural network. Thereby, the correlation between different positions can be captured. The self-attention mechanism obtains the position of each new position by calculating the distribution of self-attention of each old position and making a weighted summation. The formula of self-attention mechanism is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{\text{dim}_k}}\right) \cdot V \quad (3)$$

Q , K , V in the formula are all calculated by pre-multiplying X , the input old sequence position information, by the corresponding weight matrix W_Q , W_K , W_V .

$\text{softmax}()$ is the activation function, and dim_k calculates the dimension of the attention head.

Unlike the traditional Transformer model, whose self-attention mechanism calculates information of all positions simultaneously, the Swin Transformer uses a hierarchical structure with patch merging, allowing it to efficiently handle high-resolution images by progressively reducing spatial dimensions. The Swin Transformer network structure is shown in Fig. 3.

One core part of Swin Transformer is Swin Transformer Block. The following strategies are the core of Swin Transformer Block: windowed multi-head self-attention (W-MSA), shifted windowed multi-head self-attention (SW-MSA) and multilayer perceptron (MLP). The Swin Transformer Block structure is shown in Fig. 4.

The former makes an explanation that in the calculation of self-attention, and Swin Transformer model will not calculate the entire sequence simultaneously and directly, but limit each position to only interact with a small part of its nearby ones. Keys (K) and values (V) will also be translated in the calculation of the score of self-attention to improve the sensitivity to local information. The latter is used to do further nonlinear transformations on the features processed by the self-attention mechanism and to help the model adapt to complex requirements of feature transformation. The specific formulas for these two strategies are as follows:

$$\begin{aligned} \text{Attention}_{\text{W-MSA}}(Q, K, V) &= \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{\text{dim}_k}}\right) \cdot V \\ &\quad \text{Attention}_{\text{SW-MSA}}(Q, K, V) \\ &= \text{softmax}\left(\frac{Q \cdot (K + \text{Shift}(K))^T}{\sqrt{\text{dim}_k}}\right) \cdot V \end{aligned} \quad (4)$$

$$\text{MLP}(x) = \text{LN}(x + \text{MLP2}(\text{GELU}(\text{MLP1}(0)))) \quad (5)$$

where $\text{Shift}(K)$ is the vector obtained by performing a translation operation on K . GELU is the activation function, and MLP1 and MLP2 are two fully connected layers.

Hierarchical self-attention mechanism based on window-based attention is another core part of Swin Transformer. Swin Transformer architecture consists of multiple layers, uses patch embedding module to divide the image into nonoverlapping blocks and converts them into a set of vectors. Then, multilayer long-distance dependencies will be established between these blocks, whose window sizes are different. Generally, low-level windows are larger and used to collect overall information. As the level becomes higher, the window will gradually shrink, and focus on the process of local details. Thus, information can be effectively transferred and integrated between different layers to establish models of long-distance-dependent relations. This hierarchical

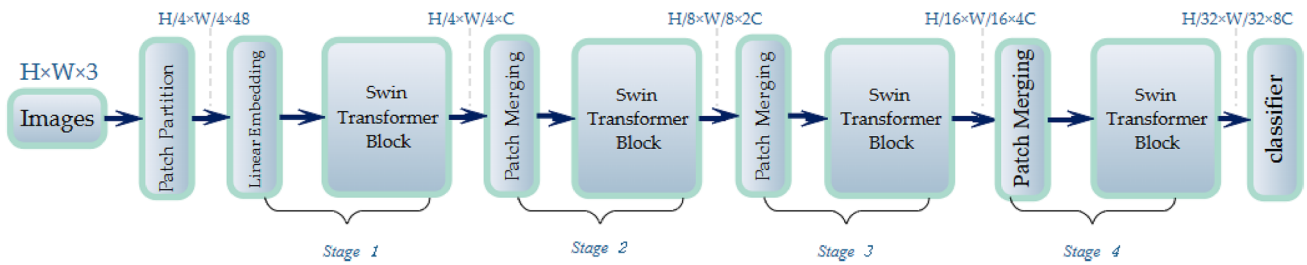


Fig. 3 Swin Transformer network structure

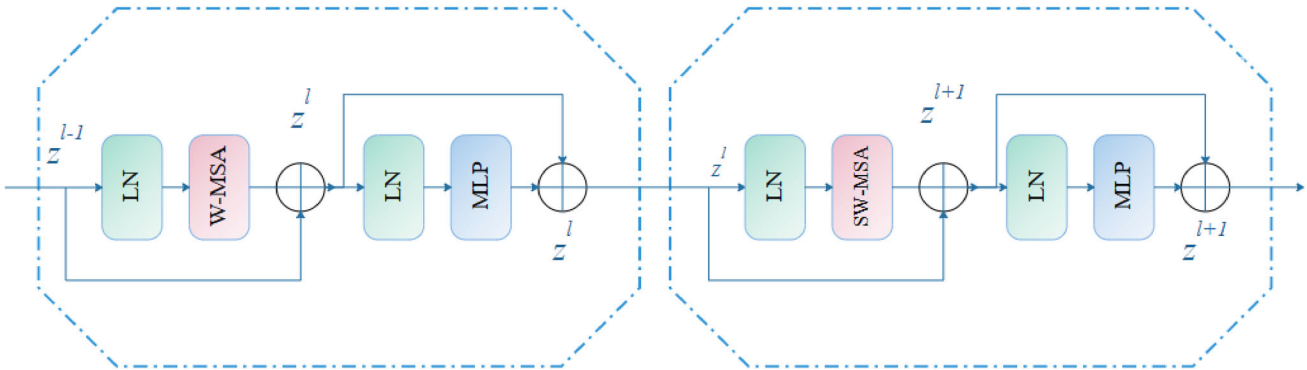


Fig. 4 Swin Transformer Block structure

self-attention mechanism allows Swin Transformer model to process large-size image data with ease, reducing computational complexity and saving memory space. Therefore, Swin Transformer has better performance when taking visual missions, such as image classification and target detection. It is also suitable for different image sizes and resolutions.

3.2 Plant disease recognition model based on CNN architecture

Convolutional neural network (CNN) [17, 18, 20, 24, 27, 28] is a deep learning model inspired by the structure and function of visual cortex of animals. It implements functions of extraction, learning and training of image features through multi-level convolution and pooling operations. CNN consists of convolutional layer (CL), pooling layer (PL) and fully connected layer (PCL). CL is the core component of CNN, which extracts features of input objects through convolution operations. The convolution kernel will perform a calculation while sliding on the input data, multiply the weight of the convolution kernel with the pixel value at the corresponding position, make a summation and finally obtain a pixel value of the output feature map. The specific calculation formula of the convolution operation is as follows:

$$B_{i,j} = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} A_{i+m,j+n} \cdot W_{m,n} + b \quad (6)$$

where B refers to the pixel value of the output feature map, k refers to the size of the convolution kernel, A refers to the feature value of the input image data, W refers to the weight of the convolution kernel and b refers to the bias function.

PL is another important layer in CNN, which is mainly used to perform the pooling operation of the image data output by the convolutional layer, reduce the spatial size of the output feature map, and extract the main features. Thus, CNN can do machine learning better. We often use max pooling (MP) and average pooling (AP) as pooling operations. The specific calculation formulas of them are as follows:

$$B_{i,j} = \max_{(x,y) \in R_{i,j}} \{A_{x,y}\} \quad (7)$$

$$B_{i,j} = \frac{1}{|R_{i,j}|} \sum_{(x,y) \in R_{i,j}} A_{x,y}$$

where $A_{x,y}$ refers to the value of the x th row and y th column of the input, $B_{i,j}$ refers to the value of the i th row and j th column of the output, $R_{i,j}$ refers to the input of the pooling area corresponding to i th row and j th column, and $|R_{i,j}|$ refers to the size of the pooling window or the number of elements. After PCL integrates the features of previous convolutional layers and pooling layers, it maps the input features to the final output category through weight matrices and bias vectors.

The learning process of the CNN model is divided into many epochs until the model can converge to or reach the predetermined stop conditions. In each epoch, the input data

will first go through a series of layers, such as convolution layers and pooling layers. Gradually, it will be transformed and expressed in high-level methods, perform forward propagation and output the prediction results of the network. In each layer, CNN learns weight parameters to gradually optimize the model. During the training process, the parameters of the model are adjusted by minimizing the loss function to reduce the gap between the model output and the true value, increasing the performance of the model. Then, the changes in the loss function are calculated through back propagation, and gradient descent algorithm is used to update the weight parameters of the model, etc. Finally, the trained model is used in the validation set and test set to verify the performance of the model on unknown datasets. CNN has made significant contributions in computer vision tasks, such as image recognition and classification.

3.3 Continual learning strategy

3.3.1 Fine-tuning strategy

Fine-tuning [43, 44] refers to using a model that has been pre-trained on a large-scale dataset, such as ImageNet as the initial model, and after fine-tuning operations, such as freezing some layers of the pre-trained model to ensure them not to participate in training. It is then applied for further training on the dataset of our experiment. Thus, the general features previously trained on ImageNet can be used to improve the performance of the model.

3.3.2 LwF strategy

Learning without forgetting (LwF) [45–47] strategy is the first deep learning algorithm to implement continual learning. It is used to solve catastrophic forgetting problem mentioned above, allowing the model to learn new tasks and reduce forgetting of previous tasks. Regularized continuous learning methods can be divided into distillation model methods and parameter importance estimation methods. LwF strategy is one of the formers based on regularization, which uses knowledge of teacher models to guide the learning of student models, helping them better learn specific tasks. Relatively, a teacher model is more complex and larger, which has been trained on other tasks and had good performance, while a student model is simpler and smaller, and needs training on a specific task.

LwF uses the output of the models in the previous stage on new data as soft labels, which will be used as additional training targets to distill the model knowledge in the previous stage, so that models in the new stage can learn the previous ones have learnt. It is equivalent to using the prediction results of previous models to guide the training of new models, making it easier to learn an output pattern similar to the previous

model, thereby minimizing the output difference between the new stage model and the previous stage model on new data, retaining knowledge of previous tasks as much as possible.

Kullback–Leibler (KL) [48] divergence is a common similarity loss function in LwF strategy. It is used to measure the difference of the output distributions between the task models in the previous stage on the new task and the in current stage. The calculation formula of KL divergence is as follows:

$$KL(P_{\text{old}} \| P_{\text{new}}) = \sum_i P_{\text{old}}(i) \log \frac{P_{\text{old}}(i)}{P_{\text{new}}(i)} \quad (8)$$

where P_{old} and P_{new} are the output probability distributions on the new task of the model trained on the previous and current ones, respectively, and i refers to the category of the output. By minimizing the KL divergence, the model can retain knowledge from previous tasks as much as possible when learning new ones. This is because during the learning process of the new task and when the output distributions of two tasks are similar, the model will be more inclined to retain the knowledge of previous tasks to reduce the impact on them.

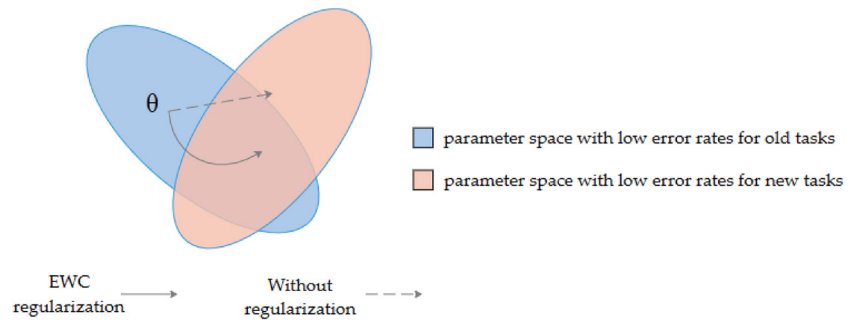
However, LwF has some requirements for limitations, such as data in new tasks and its correlation with previous ones. When the offsets of new tasks relative to previous ones are excessive, it will lead to poor performance of the model in previous tasks. Simultaneously, when more tasks are incrementally learnt, the errors made in previous tasks can gradually accumulate.

3.3.3 EWC strategy

Elastic weight consolidation (EWC) [49–51] is also a continual learning method based on regularization to solve catastrophic forgetting problem. But unlike the distillation model method such as LwF, EWC is a method of parameter importance estimation. Although both of them eventually introduce regular terms into the loss function to reduce forgetting of old knowledge of the model, LwF uses the knowledge of teacher models to guide student models learning tasks, while EWC strategy calculates relatively important parameters, and try avoid these parameters from sharp changes during the subsequent learning of new tasks. The EWC parameter update process chart is shown in Fig. 5.

EWC uses the Bayesian formula [52] to estimate the posterior distribution of parameters after joint training of the model, and uses Fisher information matrix of parameters to estimate the distribution of importance of model parameters. Fisher information matrix can reflect the effect of changes of model parameters in the loss function on the parameters. The larger the elements in its diagonal are, the more important the corresponding parameters are to previous tasks. So, in the subsequent task learning process, special attention needs

Fig. 5 EWC parameter update process chart



to be paid to avoid these important parameters from sharp changes. The formula for calculating and normalizing Fisher information matrix is as follows:

$$F_{i1} = \int \left(\frac{\partial \log p_t(D_t|\theta)}{\partial \theta_i} \right)^2 p(\theta|D_t) \quad (9)$$

$$F_{i2} = \frac{F_{i1}}{\max_k \{F_{i1}\}}$$

where F_{i1} and F_{i2} refer to the diagonal elements of the Fisher information matrix relative to the parameter θ_i before and after normalization; and $p_t(D_t|\theta)$ refers to the conditional probability of the dataset D_t under the given parameter θ . The normalized diagonal element F_{i2} will be the importance indicator of the evaluation parameter θ_i and constrain the update of the parameters in the subsequent learning process.

The main idea of the EWC strategy is to constrain the changes of some important parameters of the model by introducing a regularization term into the loss function when training new tasks. It aims to retain previous important knowledges, while allowing adaptation on new tasks. The weight of the regular terms are the diagonal elements of the previously calculated Fisher information matrix. The form of the regular term is as follows:

$$\text{Loss} = \sum_i \frac{\varepsilon F_{i2} (\theta_i - \theta'_i)^2}{2} \quad (10)$$

where ε refers to the regularization coefficient, which controls the influence of the regularization term; θ_i refers to the parameter value on the current task; and θ'_i refers to the parameter value on the previous task.

However, EWC also has limitations. For instance, it needs to calculate a Fisher information matrix for each previous task and its important distribution of each parameter. The number of regular terms will grow linearly with the number of tasks, causing a huge expense.

3.4 Federated learning strategy

Federated learning [53–56] is a distributed machine learning method, which utilizes multiple computers or devices to perform machine learning tasks. Federated learning requires each device to use local data to train the model relatively and independently, while the local data are not shared internally, and only the parameters of the locally trained model are shared with the central server. After the model parameters are transmitted to the central server, they will be organized and aggregated to update the global model parameters. Then, a repetition of the operation of local learning will be performed until the central server updates the parameters. The federated learning architecture diagram is shown in Fig. 6.

There are many aggregation operations of local model parameters by the central server, among which often used are federated averaging, weighted average aggregation and average aggregation, which is the simplest. Take federated averaging for instance, this method combines average aggregation and weighted average aggregation, whose specific formula is as follows:

$$\theta_{\text{global}} = \sum_{i=1}^N \frac{|D_i|}{|D|} \theta_i \quad (11)$$

where θ_{global} refers to the aggregated global model parameter, D_i refers to the dataset of the i th client, D refers to the sum of all client datasets and N refers to the number of clients. It can be seen that the global model parameters are the weighted average of each client parameter according to the proportion of the client data.

Federated learning method can make full use of multiple computer devices and can solve the problem of a single computer device being unable to train large datasets. Distributing tasks to multiple computers for parallel processing improves training speed and efficiency.

The use of federated learning in this task has significant advantages. Firstly, federated learning utilizes multiple computing devices to process training tasks in parallel, significantly improving the speed and efficiency of data processing and model training, and reducing the risk of

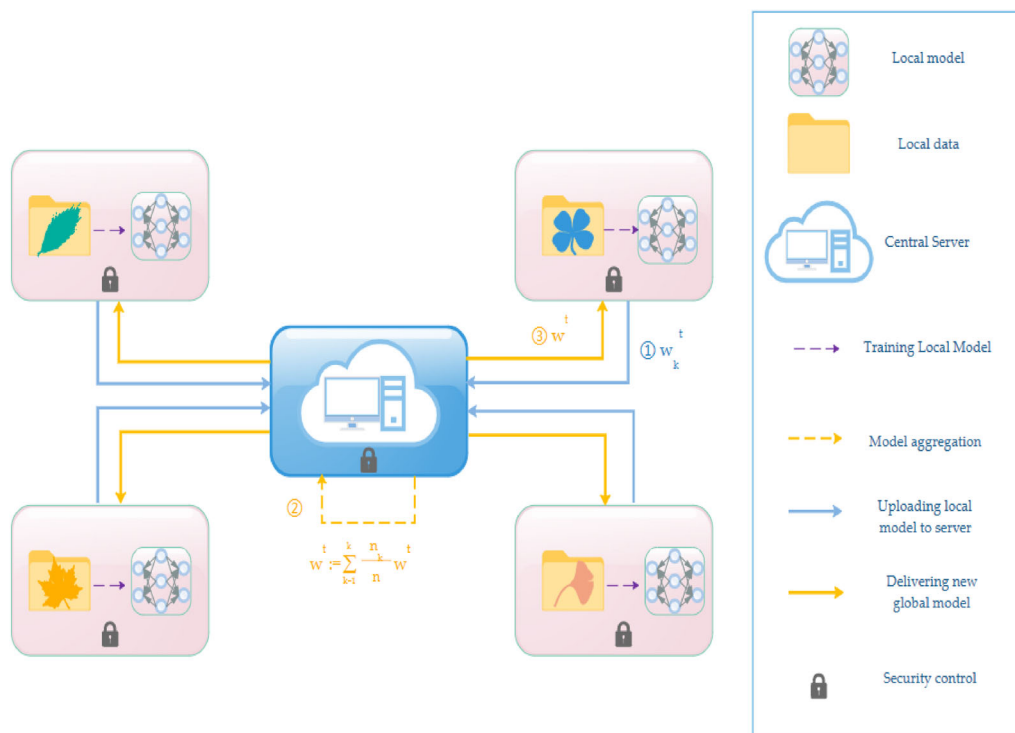


Fig. 6 Federated learning architecture diagram

single points of failure, especially for large-scale datasets. Secondly, unlike centralized learning that requires a single high-performance computing device, federated learning can use multiple ordinary performance devices for training, thus lowering hardware costs. More importantly, federated learning enables the model to adapt to continuously changing data environments, achieving higher efficiency in learning new knowledge compared to traditional centralized learning methods, thereby enhancing the model's flexibility and scalability. Additionally, agricultural data often exhibit heterogeneity, and federated learning can handle heterogeneous data by performing local training on each node, preserving the unique characteristics of each node's data and then aggregating them to form a unified model, thereby improving the overall performance of the model. Therefore, federated learning is both necessary and useful for this task, providing a more efficient and comprehensive solution for plant leaf disease identification.

3.5 *t*-SNE dimensionality reduction

t-SNE (t-distributed stochastic neighbor embedding) [57, 58] algorithm is a nonlinear dimensionality reduction technology based on SNE. It is used for mapping data from high-dimensional to low-dimensional space and form a scatter plot, according to whose boundaries of different data point groups and the cluster degree of the same one can distinguish

the quality of model recognition and classification. It can also implement the distribution status in low-dimensional spaces of data in abstract high ones. Compared with the traditional linear dimensionality reduction method PCA, benefiting from the nonlinear dimensionality reduction nature of *t*-SNE, it has better performance in generating two-dimensional or three-dimensional visualizations and displaying the features and clusters of the data. Also, given that *t*-SNE is on the basis of probability distribution to express internal relationships between data points and does not rely on specific linear structures, *t*-SNE has smaller impact on outliers and is more robust.

The working principle of *t*-SNE is to construct a probability distribution in a high-dimensional space by measuring the similarity between data points, and then construct a similar one in a low-dimensional space. To be specific, *t*-SNE first converts the high-dimensional Euclidean distance between data points into a conditional probability which represents similarity. If data point m_i in the high-dimensional space is given, the formula of the similarity conditional probability p_1 of the data point m_j is as follows:

$$p_1 = \frac{e^{(-\|m_i - m_j\|^2 / 2\sigma_i^2)}}{\sum_{k \neq i} e^{(-\|m_i - m_k\|^2 / 2\sigma_i^2)}} \quad (12)$$

where σ_i^2 refers to the Gaussian variance centered on data point m_j . If data point n_i in the low-dimensional space is

Table 2 Confusion matrix classifies the results

Confusion matrix elements	Actual	Predicted
True Positive (TP)	✓	✓
True Negative (TN)	×	×
False Positive (FP)	×	✓
False Negative (FN)	✓	×

given, the formula of the similarity conditional probability p_2 of the data point n_j is as follows:

$$p_2 = \frac{e^{(-\|n_i - n_j\|^2)}}{\sum_{k \neq i} e^{(-\|n_i - n_k\|^2)}} \quad (13)$$

Dimensionality reduction is achieved by minimizing KL divergence of the similarity probability distribution between high- and low-dimensional space. The loss function C is often calculated by minimizing KL divergence through the method of gradient descent optimization:

$$C = \sum_i \text{KL}(P_{1i} \| P_{2i}) = \sum_i \sum_j p_1 \log\left(\frac{p_1}{p_2}\right) \quad (14)$$

where P_{1i} refers to the conditional probability distributions of all other data points on the given data point m_i , while P_{2i} refers to the ones on n_i . t -SNE preserves the local structure between data points, and better displays cluster structures and their similarities. We select six network models in this article and use t -SNE algorithm to perform operations of dimensional reduction on them.

In this paper, t -SNE (t-distributed stochastic neighbor embedding) is used to perform dimensionality reduction on multiple network models, providing a visual and intuitive evaluation of these models' performance in the task of plant leaf disease recognition and classification. t -SNE excels in preserving the local structure between data points and is less affected by outliers, making it highly advantageous for generating high-quality two-dimensional or three-dimensional visualizations.

3.6 Performance metrics

Performance metrics [59] are tools to evaluate the performance of models in solving specific tasks, help us compare the performance of models, and guide us to optimize and improve them. The confusion matrix is one of the important visualization methods for comparing model performance. In Table 2, we present the four components of the confusion matrix.

3.6.1 Confusion matrix elements

3.6.2 Accuracy, precision, recall, F1 score

- *Accuracy:*

The proportion of samples for which the predictions are correct, relative to the total number of samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

- *Precision:*

The proportion of correctly predicted samples for the positive class, relative to the total number of samples predicted as positive.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

- *Recall(TPR):*

The proportion of correctly predicted samples for the positive class, relative to the total number of samples that are actually positive.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (17)$$

- *F1 Score:*

The harmonic means of precision and recall.

$$F1\text{Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (18)$$

3.7 Experimental procedure and experimental environment

The experimental workflow is illustrated in Fig. 7. It shows the steps taken from data collection to model evaluation, including data preprocessing, model training and performance evaluation.

3.8 Experimental environment

The GPU used in this experiment is an NVIDIA GeForce RTX4090 (24 GB). We used Python 3.10 (ubuntu22.04) as our programming language. The deep learning framework used is PyTorch 2.1.0 with NVIDIA CUDA 12.1 parallel computing architecture. In the experiment, we set the batch

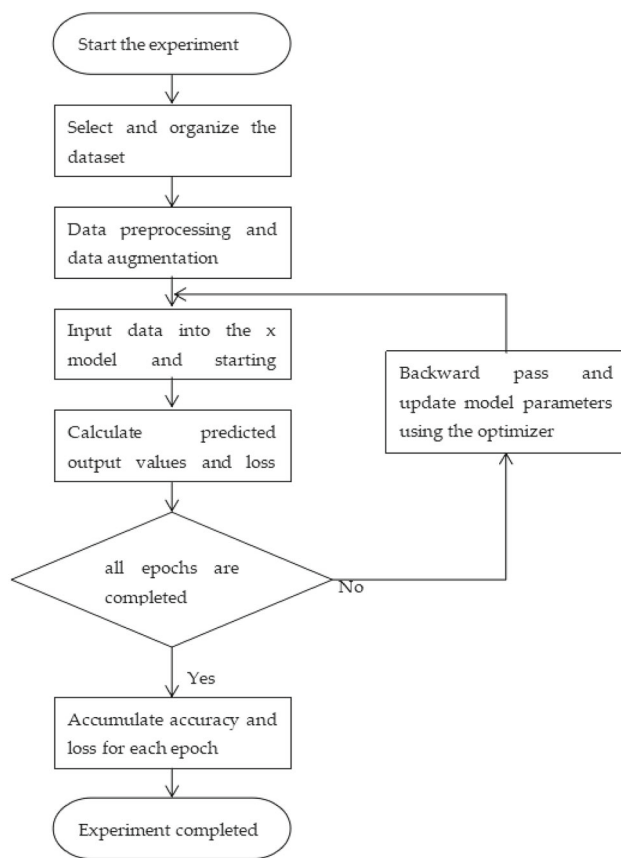


Fig. 7 Experimental workflow diagram

size to 32, the number of epochs to 50, the learning rate to $3e-5$, the gamma value to 0.7 and the seed value to 42.

4 Results

4.1 Full-batch learning

4.1.1 Comparison results of multiple models

In this experiment, we selected 10 network models, including EfficientNet-B1, MixNet-L, ResNet-18, CAIT-S24-224, GerNet_L, DenseNet-121, SelecSLS-42, LeViT-192, Poolformer-S18 (we will refer to this as PF-S18 in the following text) and Swin-Small-Patch4-Window7-224. (We will refer to this as SSPW224 in the following text.) These models are roughly divided into two architectures, Transformer and CNN. Among them, the models belonging to the Transformer architecture are CAIT-S24-224, LeViT-192, PF-S18 and SSPW224, while the models belonging to the CNN architecture are EfficientNet-B1, MixNet-L, ResNet-18, GerNet_L, DenseNet-121 and SelecSLS-42. Apparently, we can

get the conclusion that the network models with the Transformer architecture substantially perform better than those with the CNN architecture from our dataset.

We chart the training performance statistics of the above 10 models in Table 3 and sort them according to the accuracy value on the test set in ascending order. The accuracy of the model on the training set can reflect the performance of the model on given data. The higher the accuracy on the training set is, the stronger the training ability the model has. The accuracy of the model on the training set and validation set is important indicators to measure the generalization ability of the model. The model has good generalization ability if they are approximate. We can see from results that the accuracies of EfficientNet-B1 and MixNet-L are significantly lower than other models, in both training and validation sets, especially the one using Swin Transformer architecture headed by SSPW224. Simultaneously, the accuracy difference of SSPW224 between the training and test set is much smaller than other ones. So, it is not difficult to see that the training and generalization ability of SSPW224 for new knowledge are stronger than the other 9 models.

Accuracy on the test set is the most intuitive measurement to judge the ability of a model to complete the task of identification and classification of images of diseased leaves. Precision and recall, respectively, reflect its ability to accurately predict and detect all positive classes. F1-score is a comprehensive indicator of precision and recall. Model 10 has the highest accuracy, precision, recall and F1-score on the test set; in the network models of Model 6 to Model 9, there are no significant differences in accuracy and other indicators on the test set, while Model 9 is slightly stronger than others. Accuracy, precision, recall and F1-score of Model 1 to Model 5 on the test set are further behind. It illustrates from the data that SSPW224 has advantages in tasks of identification and classification of images of diseased leaves, so we use it as the basic network model.

4.1.2 Confusion matrix comparison

Confusion matrix is a kind of table often used in statistics to evaluate the performance of classification. The numbers on its diagonal represent the number of correctly classified samples in datasets of different subclasses, as TP and TN mentioned in the previous section of performance metrics. The data above the diagonal line are the number of samples that are incorrectly classified, which is TN; below is the number of samples that are incorrectly classified as this category, which is FN. We select six network models: EfficientNet-B1, MiNET-L, ResNet-18, GerNet_L, SelecSLS-42 and SSPW224. After training, we can obtain the accuracy, precision, recall and F1-score of these models on the test set and plot their confusion matrices. We sort the figures in ascending order to the accuracy of the six models on the test set.

Table 3 Comparison results of multiple models

Model Number	Name	Parameters number	Train accuracy	Valid accuracy	Test accuracy	Precision	Recall	F1 score
1	EfficientNet-B1	7,794,184	0.8580	0.9461	0.9216	0.8701	0.8790	0.8646
2	MixNet-L	7,329,252	0.8668	0.9457	0.9406	0.9099	0.9076	0.9029
3	ResNet-18	11,689,512	0.8827	0.9556	0.9531	0.9325	0.9167	0.9156
4	CAIT-S24-224	46,916,200	0.9096	0.9653	0.9585	0.9216	0.9261	0.9233
5	GerNet_L	31,078,280	0.9088	0.9617	0.9587	0.9214	0.9269	0.9238
6	DenseNet-121	7,978,856	0.9177	0.9644	0.9626	0.9323	0.9345	0.9314
7	SelecSLS-42	30,354,952	0.9360	0.9718	0.9666	0.9451	0.9404	0.9370
8	LeViT-192	10,947,069	0.9128	0.9683	0.9666	0.9436	0.9406	0.9388
9	PF-S18	56,172,520	0.9608	0.9760	0.9669	0.9432	0.9401	0.9377
10	SSPW224	49,606,258	0.9569	0.9777	0.9720	0.9525	0.9485	0.9471

The confusion matrix results for different models are shown in Fig. 8, which highlights the performance differences in identifying and classifying plant leaf diseases. There exists “sample blocks” (areas where TN and FN occur on the highest frequency) both above and below the diagonal of the confusion matrix of EfficientNet-B1. MixNet-Ls only has “sample blocks” above the diagonal, which has much fewer negative sample data points than EfficientNet-B1. ResNet-18s no longer have “sample blocks,” but remains an enormous number of scattered negative sample data points above and below the diagonal. The confusion matrices of GerNet_L and SelecSLS-42 are roughly similar. Compared with ResNet-18, they have much fewer negative sample data points. The samples of SSPW224 are roughly adjacent to the diagonal of the confusion matrix, and only a few negative sample points are in other areas. Thus, the TP and FP of SSPW224 are significantly higher than other models, while TN and FN are lower. Therefore, it can be seen from the results of the confusion matrices of the 10 models in the experiment that the SSPW224 model has the optimal performance in the task of identification and classification of images of diseased leaves.

4.1.3 Performance results of the SSPW224 model on subclasses

What is reflected in the Table 4 is precision, recall, F1-score and the number of image samples in each subclass dataset of diseases of the SSPW224 model in the task of identification and classification in them. Our experiment integrates other datasets based on the original one. And we have endeavored to ensure an ample number of samples in the test set, with the majority of test set samples exceeding 200 in quantity. It can be seen that the SSPW224 model is effective in the classification of diseases of leaves of every plant, except for the classification of diseases of rice. We speculate that it is probably because the leaves of rice are elongated shapes

and pointed at both ends, so there are not many identifiable featured areas on the leaves. The features of the leaves of rice brown spot, rice healthy, rice Hispa and rice leaf blast are similar, making it difficult for the models to distinguish, and the precision, recall and F1-score of the SSPW224 model in the classification of different subclasses of diseases relatively low.

4.1.4 Model comparison after *t*-SNE dimensionality reduction

Confusion matrices are a common method for visualization. Next, we will use *t*-SNE dimensionality reduction as a supplementary visualization. *t*-SNE can clearly show the distribution and clustering of data points, effectively evaluating the performance of the six selected models in plant leaf disease recognition and classification tasks.

In the dimensionality reduction operation, we set the number of dimensions of the reduced feature space (*n_components*) as 2, the early exaggeration factor (early exaggeration) which controls *t*-SNE as 3, the random number generator in the initializer of *t*-SNE (*random_state*) as 8, perplexity as 7 and the learning rate (*learning_rate*) as 300. Figure 9 shows that the boundaries of the classification of data points become clearer gradually, and the distribution is more concentrated. Therefore, the SSPW224 model is optimal. Initially, the EfficientNet-B1 model has 1280 features, the MixNet-L model has 1536 features, the ResNet-18 model has 512 features, the GerNet_L model has 2560 features, the SelecSLS-42 model has 2048 features, and the Swin-Small-Patch4-Window7-224 model has 768 features. All of these features are reduced to 2 dimensions using *t*-SNE.

Figure 9 illustrates the clustering results of different models after *t*-SNE dimensionality reduction, clearly showing the differences in clustering performance among the models. Taking the green and orange data points in the lower

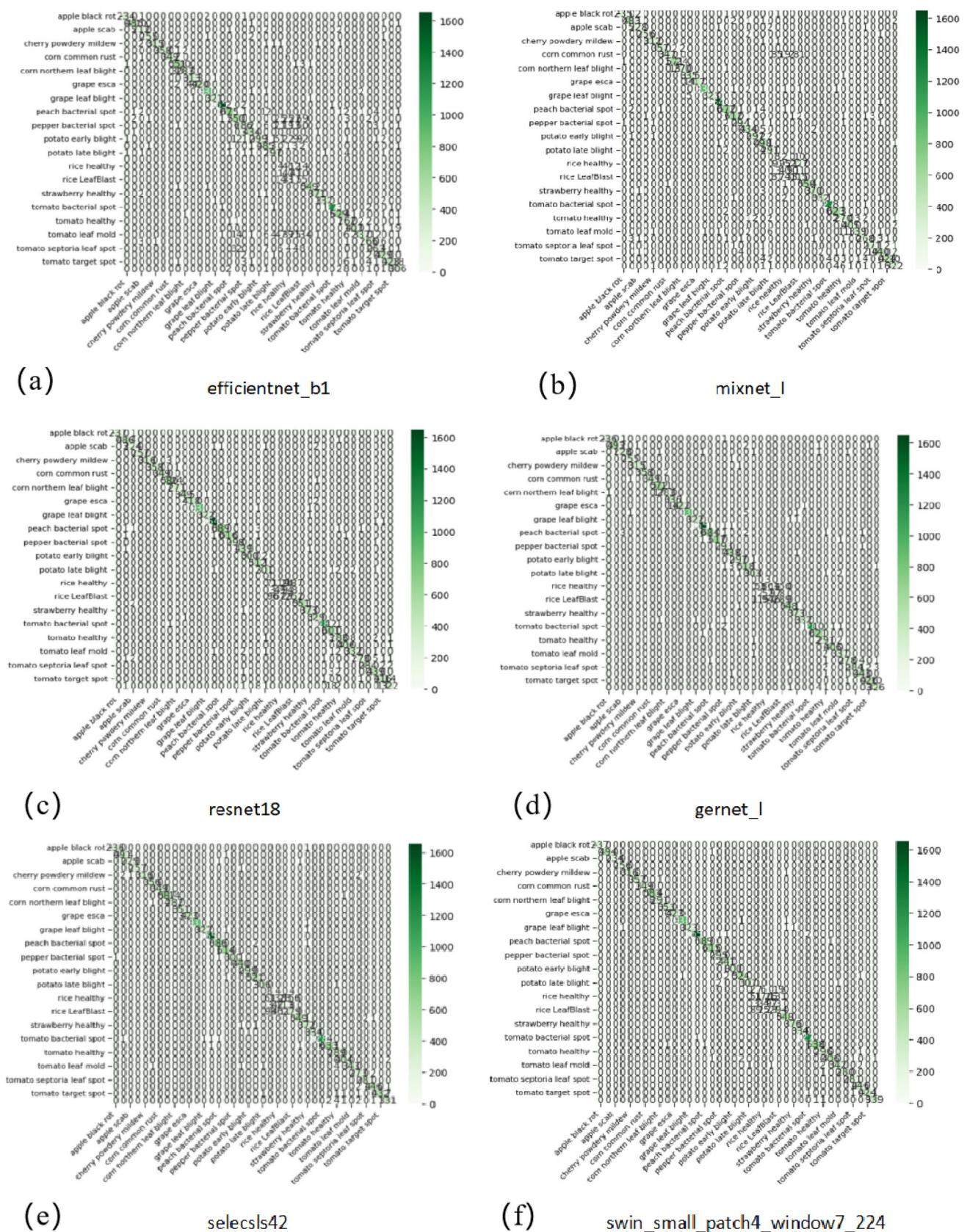
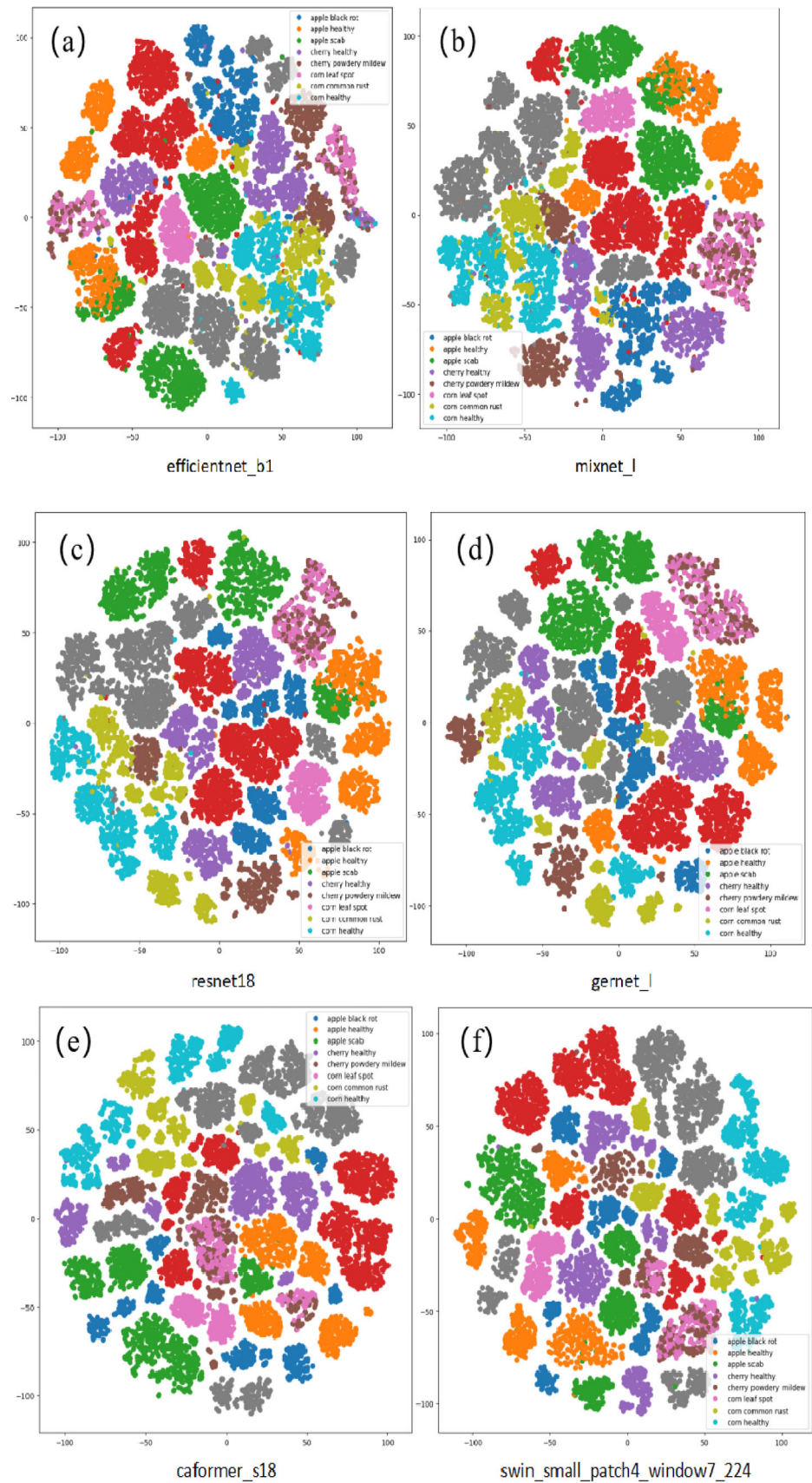


Fig. 8 Confusion matrix comparison results, **a** efficient_b1, **b** mixnet_i, **c** resnet18, **d** gernet_i, **e** selecsls42, **f** swin_small_patch4_window7_224

Table 4 Performance results of the SSPW224 model on subclasses

Disease category	Precision	Recall	F1-score	Support
Apple black rot	1.0000	1.0000	1.0000	237
Apple healthy	0.9980	1.0000	0.9990	494
Apple scab	1.0000	1.0000	1.0000	234
Cherry healthy	1.0000	0.9961	0.9981	257
Cherry powdery mildew	0.9968	1.0000	0.9984	316
Corn leaf spot	0.9972	0.9972	0.9972	358
Corn common rust	0.9971	1.0000	0.9986	349
Corn healthy	0.9932	0.9865	0.9898	591
Corn northern leaf blight	0.9732	0.9831	0.9782	296
Grape black rot	1.0000	0.9972	0.9986	354
Grape esca	1.0000	1.0000	1.0000	423
Grape healthy	0.9979	0.9989	0.9984	935
Grape leaf blight	0.9908	1.0000	0.9954	323
Orange HLB	0.9976	0.9994	0.9985	1653
Peach bacterial spot	0.9986	0.9986	0.9986	690
Peach healthy	1.0000	0.9968	0.9984	617
Pepper bacterial spot	1.0000	0.9833	0.9916	300
Pepper healthy	0.9955	0.9932	0.9944	444
Potato early blight	1.0000	1.0000	1.0000	300
Potato healthy	0.9924	0.9943	0.9934	527
Potato late blight	0.9903	0.9968	0.9935	308
Rice brown spot	0.5192	0.1588	0.2432	147
Rice healthy	0.4497	0.7234	0.5546	447
Rice Hispa	0.7080	0.6599	0.6831	170
Rice leaf blast	0.6821	0.6577	0.6697	235
Squash powdery mildew	1.0000	0.9946	0.9973	551
Strawberry healthy	1.0000	0.9973	0.9987	377
Strawberry leaf scorch	1.0000	1.0000	1.0000	334
Tomato bacterial spot	0.9992	0.9969	0.9980	1283
Tomato early blight	0.9953	0.9984	0.9969	639
Tomato healthy	0.9966	0.9867	0.9916	300
Tomato late blight	0.9878	0.9975	0.9927	407
Tomato leaf mold	0.9828	0.9971	0.9899	343
Tomato mosaic virus	0.9964	1.0000	0.9982	280
Tomato septoria leaf spot	1.0000	0.9894	0.9947	284
Tomato spider mites	0.9978	0.9911	0.9944	450
Tomato target spot	0.9930	0.9792	0.9860	433
Tomato TYLCV	0.9686	0.9941	0.9812	341
Accuracy			0.9720	17,027
Macro avg	0.9525	0.9485	0.9471	17,027
Weighted avg	0.9728	0.9720	0.9710	17,027

Fig. 9 Comparison of the clustering results of the six models after t -SNE dimensionality reduction: **a** efficientnet_b1, **b** mixnet_i, **c** resnet18, **d** gernet_i, **e** caformer_s18, **f** swin_small_patch4_window7_224



left corner of the first figure for instance, in the EfficientNet-B1 model, the data points of the two colors are inseparable. In MixNet-L, ResNet-18 and GerNet_L models, these data points move to the upper right but are still clustered together without clear boundaries, though the separation is slightly better than in the EfficientNet-B1 model. In the SelecSLS-42 model, this group of data points has moved to the top. The distribution of data points in the two colors is relatively clear, but there is still no obvious boundary separating them. In the optimal model, SSPW224, this group moves to the middle and a clear boundary appears, with the data points in different colors being more distinctly clustered than in the other models. Thus far, we assume that SSPW224 is the best algorithm among the ten, based on the scatter plot obtained by *t*-SNE dimensionality reduction, as it shows the most distinct and concentrated clusters.

4.2 Federated continual learning

4.2.1 Integration of two models with federated continual learning

It can be seen from the experiment results in the “full-batch learning” part that SSPW224 and PF-S18 are the two network models which, respectively, rank first and second in general performance among all models. So, we apply three continuous learning strategies (fine-tuning, EWC and LwF) to two network models, SSPW224 and PF-S18, and add federated learning strategies with different numbers of federated users to each strategy model, forming 18 different combinations. The experimental results are shown in Table 5 and Table 6 respectively. It can be seen from the experiment results that SSPW224 and PF-S18 models perform significantly better than other ones in average accuracy when adding LwF deep learning strategy and setting the number of learning federated users as 3. Average accuracy reflects the average performance of the model throughout the continuous learning process. We can assert that the average accuracy of SSPW224 is higher than the one under the corresponding same strategy in PF-S18, which verifies that the SSPW224 model has a better performance on our dataset, better abilities of generalization and feature extraction, more accuracy in predictions on unseen data.

Previous and current, respectively, refer to the accuracy of old and new data, which can reflect the adaptability of the model to new ones. It can be seen from the experiment results that the SSPW224 model has much higher accuracy than the PF-S18 model on new data under combinations of different strategies, so the generalization ability of SSPW224 is significantly stronger, making it easier to accept new trains. PF-S18 will be more likely to encounter overfitting problems. In addition to SSPW224, except from the aggregated LwF strategy, new data in the remaining six strategy combinations

have higher accuracy than the old ones. In contrast, they have similar accuracy in the combinations of the LwF strategy. For PF-S18, except for aggregated fine-tuning strategy and the experiment which has 5 federated users, in the other eight groups, new data are significantly lower in accuracy than old ones. This demonstrates that the SSPW224 model not only has better stability and good performance on different datasets, but also has better performances on new data and stronger ability in learning.

4.2.2 Visualization of federated continual learning

The comparison of various model strategy combinations with different numbers of federated users is shown in Fig. 10. It highlights the accuracy changes over training rounds for the SSPW224 and PF-S18 models with federated continual learning strategies. The three figures above are listed in order of 3, 5 and 8 by the number of federated users, which show the trends of the top 1 accuracy on data of test sets with training rounds for two models aggregated with different continuous learning strategies. In each figure, the decline of the color curve that represents the SSPW224 model aggregated with LwF strategy is optimal, almost higher at every node than others. This represents the SSPW224 model aggregated with the EWC strategy has slightly less decline than the former in each figure, but is still better than the other 4 strategies. When descending, the curve represents the SSPW224 model aggregated with the fine-tuning strategy and has a high frequency of crosses with the one that represents the PF-S18 model combined with the LwF strategy. But at positions of final nodes, the former always has higher accuracy than the latter. So generally speaking, the ability of generalization of the former is better than the latter. Therefore, the performance of the SSPW224 model combined with any federated continuous learning strategy is better than the PF-S18 models.

By comparing the declining range of the color curve which refers to the SSPW224 model in aggregation with LwF policy in three figures, it can be seen that when the number of federated users is 3, the curve drops at the slowest rate and its final node is higher when the number is 5 or 8. It is equivalent to the situation that in the last round of experiments, the former is higher in accuracy rate than the latter when it has the strongest ability to learn new things. Thus, we conclude from two perspectives of data and visualization that the addition of LwF policy to the SSPW224 model, while setting the number of learning federated users to 3 (which abbreviation is SSPW224-LwF-3) is the optimal strategy for this experiment.

Table 5 Integration of SSPW224 models with federated continual learning

Model	Deep learning strategy	Number of users	Previous (%)	Current (%)	Average accuracy (%)
SSPW224	Fine-tuning	3	7.13	98.87	20.95
		5	17.14	88.61	27.90
		8	16.15	84.63	26.46
	EWC	3	23.68	71.92	30.94
		5	17.16	86.04	27.53
		8	21.53	80.81	30.46
	LwF	3	49.41	33.81	47.06
		5	43.04	36.51	42.06
		8	34.33	22.82	32.59

Table 6 Integration of PF-S18 models with federated continual learning

Model	Deep learning strategy	Number of users	Previous (%)	Current (%)	Average accuracy (%)
PF-S18	Fine-tuning	3	11.57	7.18	10.91
		5	9.14	25.78	11.65
		8	9.93	5.62	9.28
	EWC	3	11.46	4.72	10.45
		5	14.61	0.27	12.45
		8	9.65	4.60	8.89
	LwF	3	22.35	6.55	19.97
		5	20.46	1.72	17.64
		8	20.67	1.4	17.77

5 Discussion

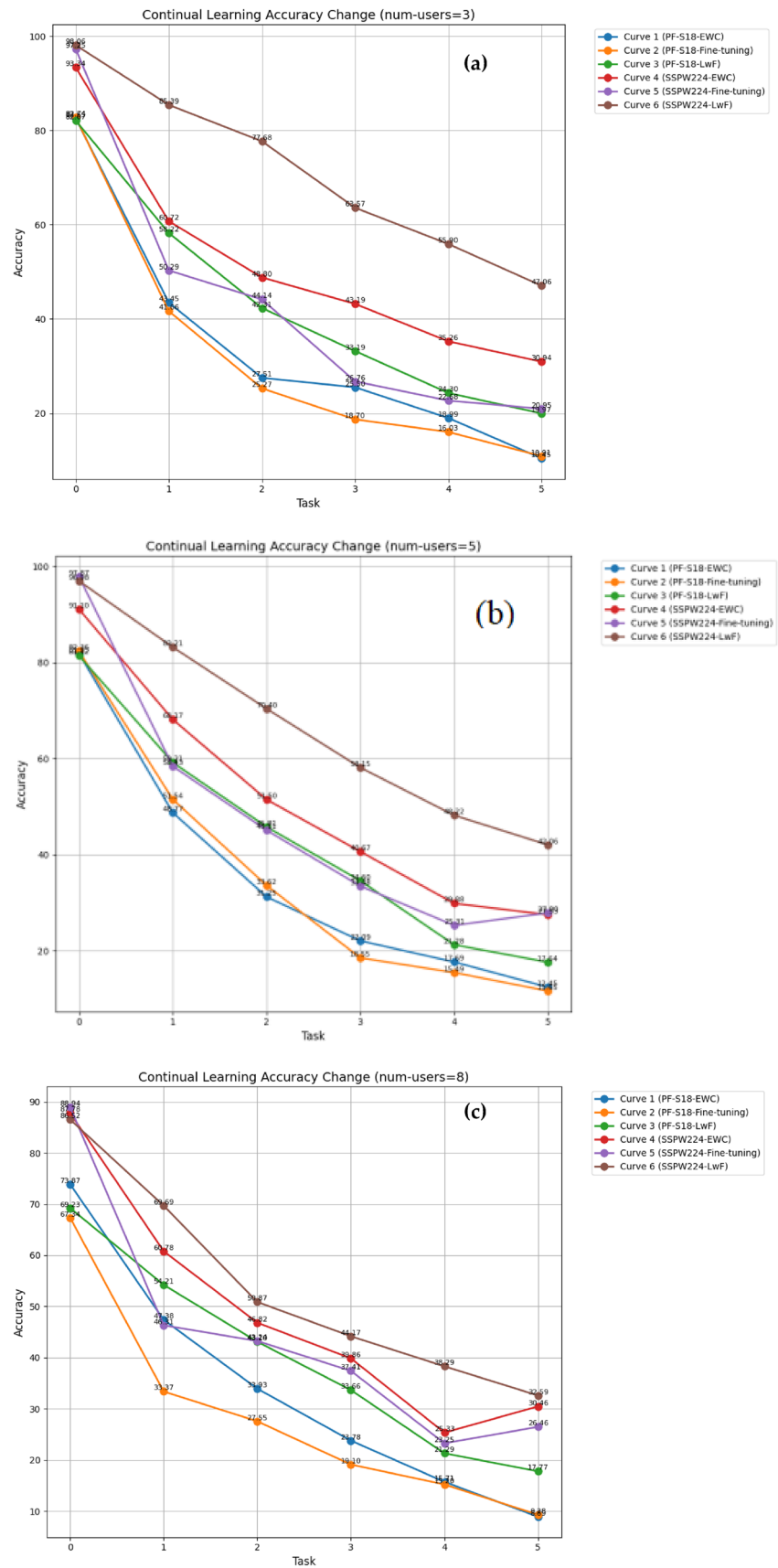
In this work, we propose a model based on the advanced Swin Transformer architecture, combined with federated learning and continual learning strategies. This model possesses the capabilities of continuous incremental learning and distributed federated learning, specifically designed for identifying and classifying plant leaf diseases. Traditional machine learning and deep learning models have achieved some success in identifying plant leaf diseases, but they still face several issues. For instance, traditional models are limited by the complexity and scale of the data, making it difficult to handle large and complex datasets. Additionally, neural networks suffer from catastrophic forgetting, where the performance on previous tasks degrades when learning new tasks, making it challenging for existing models to maintain good performance on both new and old tasks simultaneously.

To address the limitations of traditional models, recent research efforts have made significant advancements. Sunil, C.K. and colleagues achieved high accuracy in plant disease detection using the EfficientNetV2 model, which excels in processing complex image data [60]. Their systematic study also explored deep learning methods for plant disease classification, proposing an ensemble approach with

AlexNet, ResNet50 and VGG16. These studies demonstrated superior accuracy in disease detection [61, 62]. Additionally, they focused on specific plant diseases (tomato and cardamom), enhancing the effectiveness of deep learning models in recognizing single diseases [60, 63]. Our research has also addressed and improved upon traditional issues. The model proposed in our study is based on the advanced Swin Transformer architecture, which incorporates a local attention mechanism, enhancing the ability to extract detailed features, particularly when dealing with complex and large datasets. Moreover, previous research in the field of plant leaf disease identification has lacked solutions to issues such as catastrophic forgetting and parallel learning. The model proposed in this study integrates continual learning strategies and federated learning strategies, enabling it to handle distributed training tasks and enhancing the model's ability to adapt to new environments.

First of all, we established a comprehensive dataset of plant leaf disease, which includes images of various plant leaf diseases and provides abundant samples for the training and validation of our model. It also simplifies the work for future researchers, who can directly use our well-organized dataset without the need for reorganization. Afterward in

Fig. 10 Comparison of various model strategy combinations with different numbers of federated users, **a** continual learning accuracy change (num-users = 3), **b** continual learning accuracy change (num-users = 5) and **c** continual learning accuracy change (num-users = 8)



the data preprocessing, we applied data augmentation techniques to the dataset such as pretzel noise, Gaussian noise and blurring the image. After that, we selected 10 network models based on Transformer architecture and CNN architecture. We evaluated the different models in terms of analyzing the performance metrics and visualizing the *t*-SNE dimensionality reduction, and finally selected the best-performing model SSPW224 and the second-best-performing model PF-S18 as the base network models. Both models are based on the Transformer architecture, and in particular, the best-performing SSPW224 model is based on the Swin Transformer architecture which is improved according to the Transformer architecture. It is designed with a local attention-based mechanism, which splits the input image into different patches for processing, instead of directly handling the whole image. Compared to the conventional CNN architecture, this patch-based processing reduces the number of parameters to be trained in the model, so it delivers higher efficiency in processing images. This reflects the necessity to enhance the capability of detailed feature extraction of this base network model for better identification and classification of plants similar to rice, where no obvious disease traits can be observed on the leaves. After that, we added continuous learning strategies such as fine-tuning, LwF and EWC and the strategy of federated learning to the base network model, and set up a total of three different numbers of federated users in the federated learning system, consisting of 3, 5 and 8. Finally, the optimal combination SSPW224-LwF-3 was selected from 18 different combinations of model strategies. By comparing the training results of different models and federated continuous learning strategies on a unified dataset, the SSPW224-LwF-3 proposed in this paper outperformed other combinations of network models and federated continuous learning strategies in both performance and recognition.

In this paper, most of the plants in our selected dataset are food crops. The normal productivity of food crops is of its importance to the world's food security and economic development [64]. The final model combines strong performance in classifying plant leaf diseases with continuous learning and federated distributed learning capabilities. The model can be utilized in intelligent agriculture [65, 66] to automate the process of identifying whether a plant has a disease and what kind of disease it has. The continuous learning capability allows the model to rapidly adapt to new environments and learn new knowledge so that it can be applied according to the characteristics and needs of different regions and farms, as well as continuously update its domain knowledge of leaf diseases. The competency of federal learning reduces the performance requirements of computing devices. Plenty of organizations or individuals working in agriculture do not have high-performance computers, and federated learning helps to bridge this gap. Also, training local datasets from different regions and sharing parameters facilitate the model

to learn different diseases of different plants in multiple geographic regions. Even in the face of exotic species or foreign diseases in the future, it can also provide early warning and promote the increase of agricultural informatization level.

6 Conclusions

In the field of identifying plant diseases, the correct identification and classification of plant diseases are the fundamental prerequisite for targeted treatment when we face a wide range of plant diseases. Our study proposes a federated continual learning model (SSPW224-LwF-3), which not only inherits the high performance of the basic network model in the field of identifying and classifying plant diseases but also gets the ability for continual learning and distributed federated learning. Firstly, through the experimental results presented in Sect. 4.1, it can be observed that in the full-batch learning experiments, the SSPW224 model outperforms other network models in the identification and classification of plant leaf diseases, exhibiting the highest accuracy and robustness among the models tested. The second-ranked model PF-S18 had lower test set accuracy, precision, recall and F1-score of 96.69%, 94.32%, 94.01% and 93.77% on the same datasets, slightly inferior to the SSPW224 model. After incorporating federated continual learning strategies, the average accuracy of the SSPW224-LwF-3 model strategy combination reached the highest at 47.06% on the same datasets. And its accuracies for old and new data, respectively, reach 49.41% and 33.81%. This comprehensive performance outperformed 17 other model strategy combinations. The results indicate that the comprehensive model proposed in this study (SSPW224-LwF-3) has achieved excellent and accurate results in detecting plant leaf diseases, surpassing other classical network models.

However, there are still some urgent issues that need to be addressed. The experimental data demonstrated that our models may have degraded performance in the face of leaf diseases that have highly similar characteristics and are not supported by a large amount of data. Shortly, we will establish a larger dataset to continue to optimize these models, while adding more variety and quantities of datasets to train and improve the models, so that they can still maintain high accuracy in more complex datasets, and detect subtle differences in pathology symptoms in the face of similar plant features. Eventually, the models can be utilized in the development of smart agriculture, contributing to global food security.

Author contributions H.Z. was involved in conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing-original draft preparation, visualization and project administration. H.Z. and G.R. were responsible for writing-review and editing, and supervision. All authors have read and agreed to the published version of the manuscript.

Data availability Relevant data are made available in open repository. Database link: https://osf.io/v4qfr/?view_only=38b53e39988c4e1e82363031a921f799.

Declarations

Conflicts of interest The authors declare no competing interests.

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